

# AI Calling Platform - GSM, Asterisk & FreeSWITCH Architecture

This document explains how an enterprise AI calling platform connects GSM gateways, Asterisk/FreeSWITCH and the AI engine.

## Overall Flow

Company Campaign → Queue → Telephony Engine → Asterisk → GSM Gateway → Physical SIM → Telecom Network → Customer. Customer audio returns to Asterisk, then Faster Whisper converts speech to text, Ollama generates the reply, Kokoro XTTS creates speech, and Asterisk sends the audio back through the GSM gateway and SIM.

## GSM Gateway

A GSM Gateway is hardware that contains SIM cards and converts SIP/VoIP calls into GSM mobile calls. Typical devices include GOIP, Dinstar and OpenVox.

## Asterisk vs FreeSWITCH

Asterisk is easier to integrate, has excellent AMI/ARI APIs, works well with GSM gateways and is ideal for outbound AI calling. FreeSWITCH offers stronger media processing and very high scalability but is generally more complex.

## Recommended Choice

For this project use Asterisk because it provides reliable outbound calling, recording, SIP integration, GSM gateway compatibility and simpler AI integration.

## Required Components

NestJS Backend, React Frontend, MySQL + Prisma, Redis + BullMQ, Asterisk Server, GSM Gateway, Physical SIMs, Faster Whisper, Ollama and Kokoro XTTS.

## AI Call Lifecycle

Campaign starts → Queue selects contact → Asterisk dials → GSM Gateway uses SIM → Customer answers → Audio streams to AI → Whisper STT → Ollama → XTTS → Customer hears AI → Call recording and analytics saved.

## Conclusion

The recommended production architecture is: React → NestJS → BullMQ → Asterisk → GSM Gateway → Physical SIM → Customer, with Faster Whisper, Ollama and Kokoro XTTS connected to Asterisk for real-time conversation.